RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing

Object Oriented Programming Lab

Lab 2

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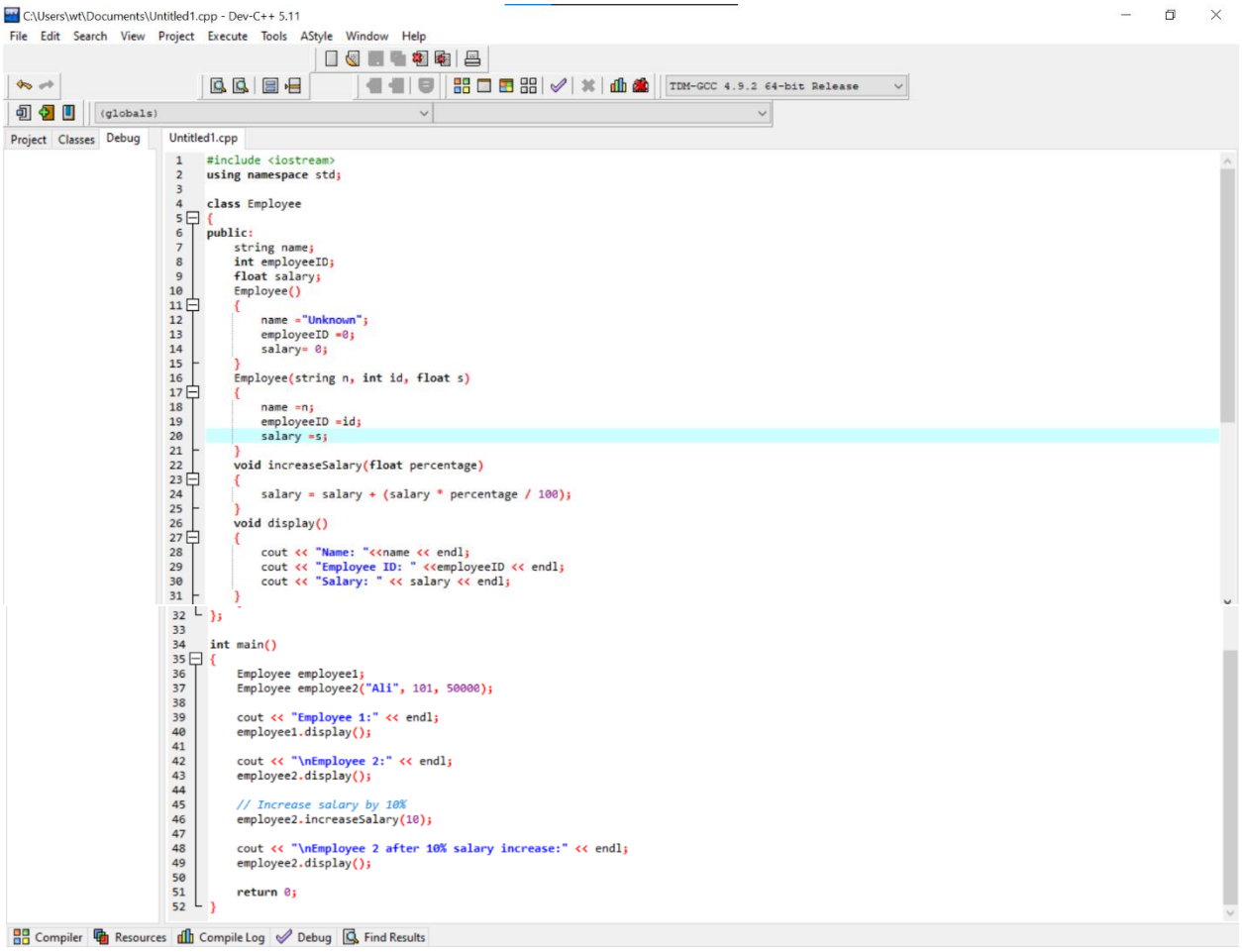
Faculty of Computing

Lab Tasks

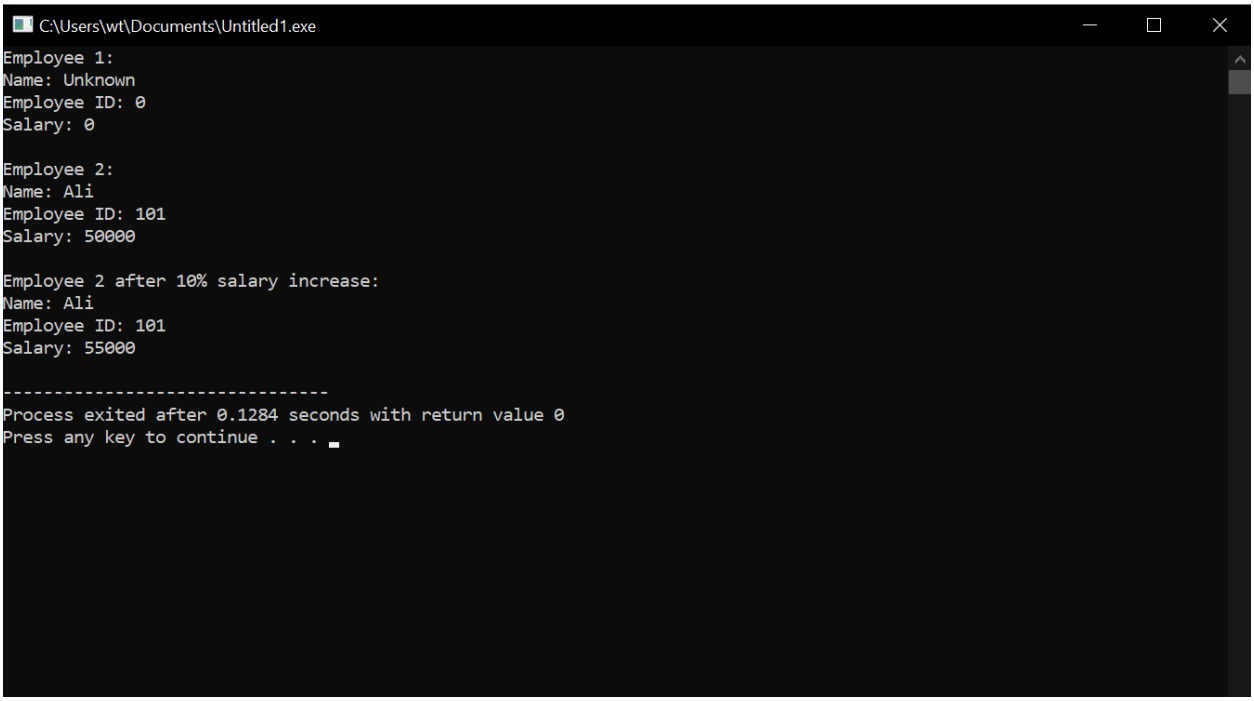
Task 1 — Employee Records

Create an Employee class with private members name, employeeID and salary. Implement both a default constructor and a parameterized constructor. Create two objects and display their information.

Extension: Add a function that increases salary by a given percentage.



Output:

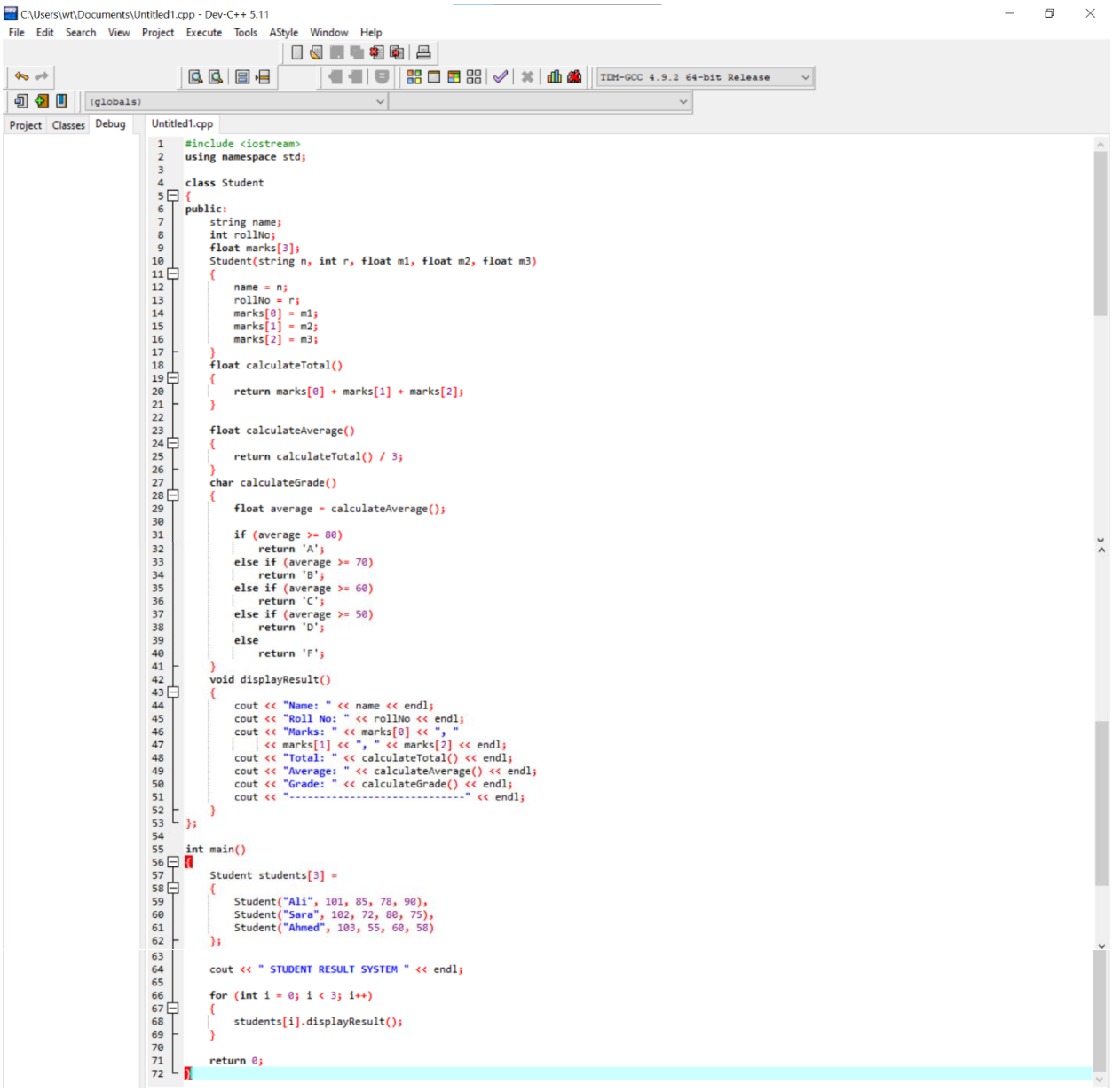


Task 2 — Student Result System

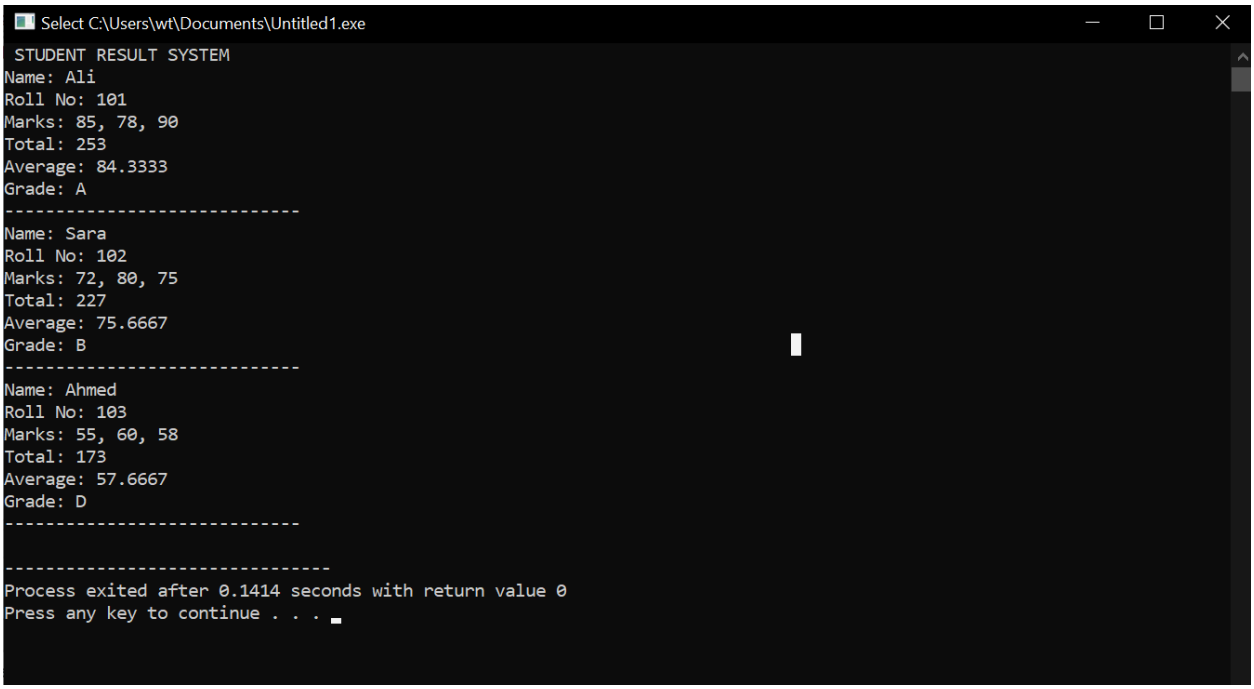
Design a Student Result System using a class named Student. Store public members name, rollNo and marks for three subjects. Use a parameterized constructor and member functions to calculate total, average and grade. Create an array of at least three Student objects and display the result of every student.

Requirements:

Use public data members.
Use a constructor to initialize each student.
Use separate functions for total and average.
Assign a grade based on average.
Use an array of objects.
Display a neat result for all students.

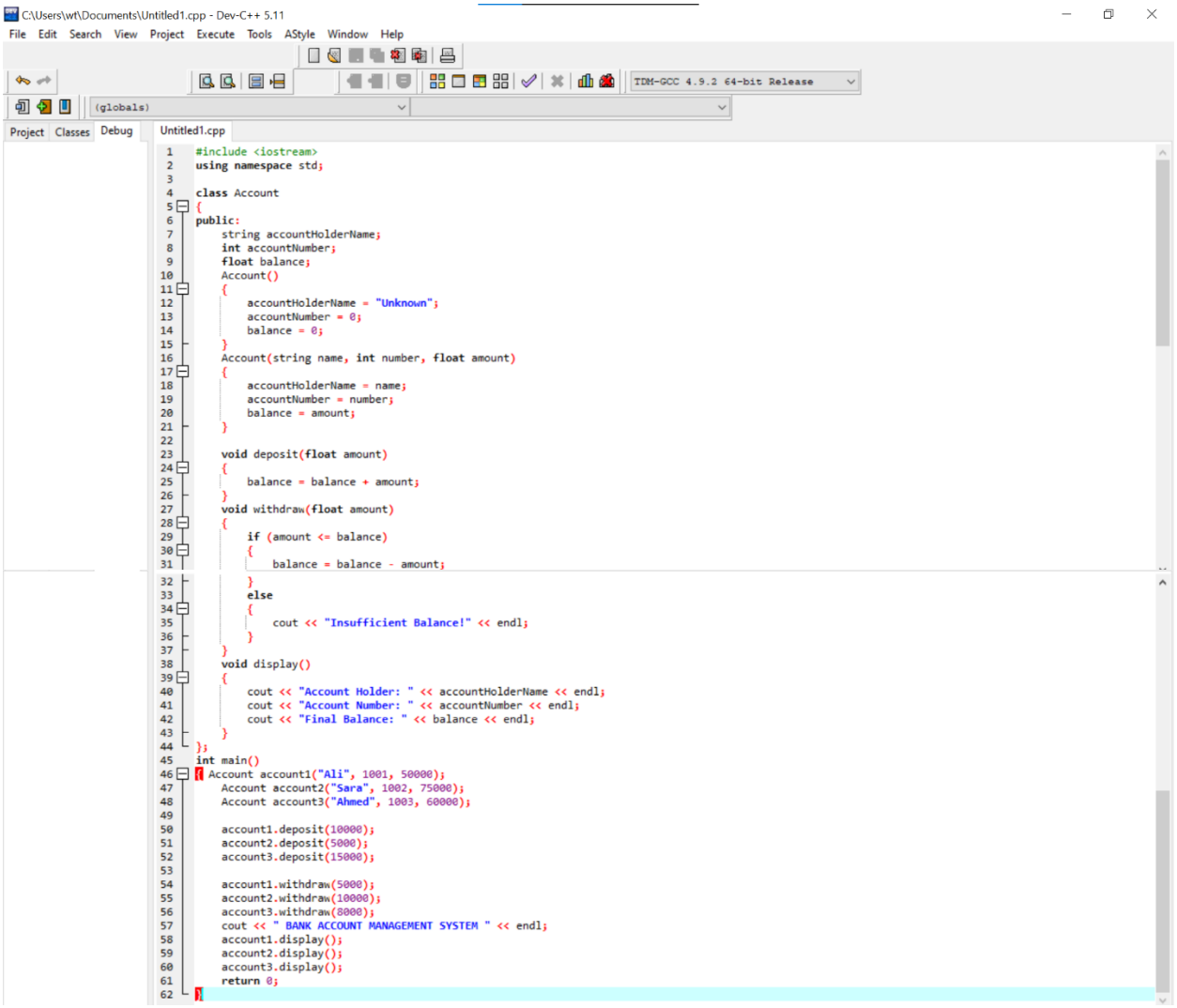

C:\Users\wt\Documents\Untitled1.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
Untitled1.cpp
#include <iostream>
using namespace std;
class Student
{
public:
string name;
int rollNo;
float marks[3];
Student(string n, int r, float m1, float m2, float m3)
{
name = n;
rollNo = r;
marks[0] = m1;
marks[1] = m2;
marks[2] = m3;
}
float calculateTotal()
{
return marks[0] + marks[1] + marks[2];
}
float calculateAverage()
{
return calculateTotal() / 3;
}
char calculateGrade()
{
float average = calculateAverage();
if (average >= 80)
return 'A';
else if (average >= 70)
return 'B';
else if (average >= 60)
return 'C';
else if (average >= 50)
return 'D';
else
return 'F';
}
void displayResult()
{
cout << "Name: " << name << endl;
cout << "Roll No: " << rollNo << endl;
cout << "Marks: " << marks[0] << ", "
<< marks[1] << ", " << marks[2] << endl;
cout << "Total: " << calculateTotal() << endl;
cout << "Average: " << calculateAverage() << endl;
cout << "Grade: " << calculateGrade() << endl;
cout << "------------------------------" << endl;
}
};
int main()
{
Student students[3] =
{
Student("Ali", 101, 85, 78, 90),
Student("Sara", 102, 72, 80, 75),
Student("Ahmed", 103, 55, 60, 58)
};
cout << " STUDENT RESULT SYSTEM " << endl;
for (int i = 0; i < 3; i++)
{
students[i].displayResult();
}
return 0;
}

Output:

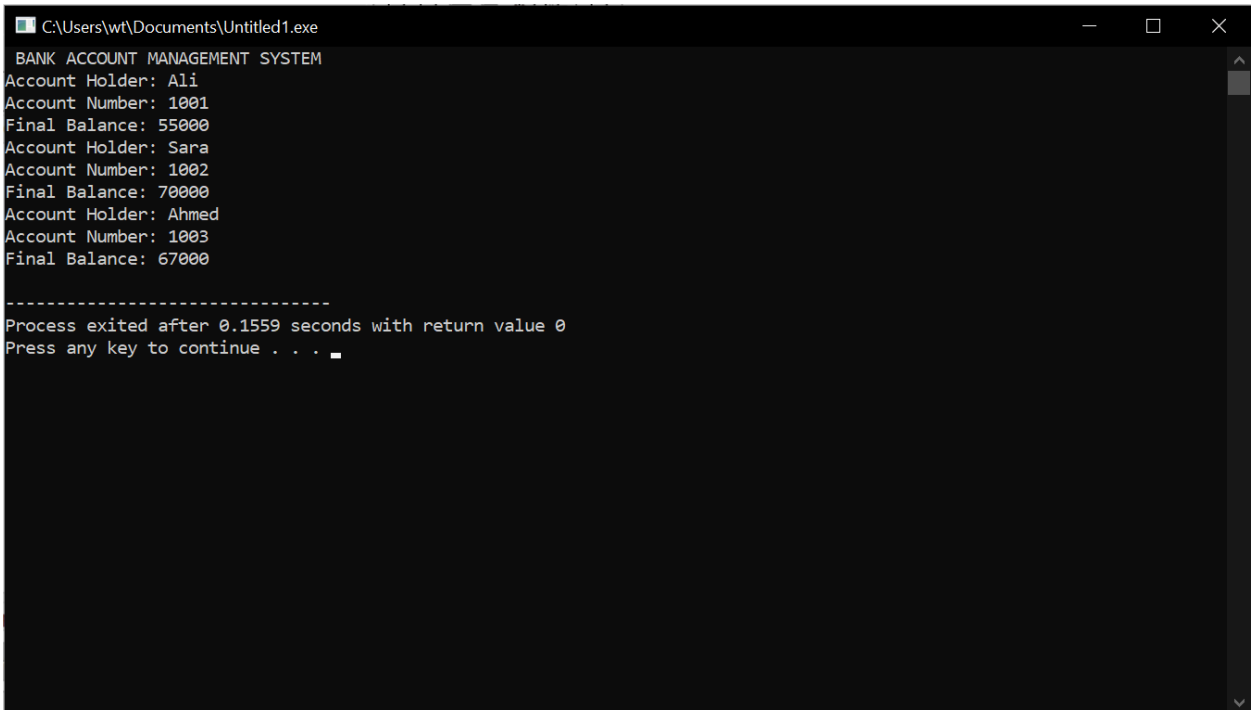


Task 3 — Bank Account Management System

- Create an Account class with:
  - Account holder name, account number, and balance.
  - Use appropriate access specifiers.
  - Implement a default constructor and a parameterized constructor.
  - Create objects for 3 accounts.
  - Create functions to deposit, withdraw, and display account information.
  - Display the final balance of each account.


C:\Users\wt\Documents\Untitled1.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
Untitled1.cpp
#include <iostream>
using namespace std;
class Account
{
public:
string accountHolderName;
int accountNumber;
float balance;
Account()
{
accountHolderName = "Unknown";
accountNumber = 0;
balance = 0;
}
Account(string name, int number, float amount)
{
accountHolderName = name;
accountNumber = number;
balance = amount;
}
void deposit(float amount)
{
balance = balance + amount;
}
void withdraw(float amount)
{
if (amount <= balance)
{
balance = balance - amount;
}
else
{
cout << "Insufficient Balance!" << endl;
}
}
void display()
{
cout << "Account Holder: " << accountHolderName << endl;
cout << "Account Number: " << accountNumber << endl;
cout << "Final Balance: " << balance << endl;
}
};
int main()
{ Account account1("Ali", 1001, 50000);
Account account2("Sara", 1002, 75000);
Account account3("Ahmed", 1003, 60000);
account1.deposit(10000);
account2.deposit(5000);
account3.deposit(15000);
account1.withdraw(5000);
account2.withdraw(10000);
account3.withdraw(8000);
cout << " BANK ACCOUNT MANAGEMENT SYSTEM " << endl;
account1.display();
account2.display();
account3.display();
return 0;
}

Output:


C:\Users\wt\Documents\Untitled1.exe
BANK ACCOUNT MANAGEMENT SYSTEM
Account Holder: Ali
Account Number: 1001
Final Balance: 55000
Account Holder: Sara
Account Number: 1002
Final Balance: 70000
Account Holder: Ahmed
Account Number: 1003
Final Balance: 67000

Process exited after 0.1559 seconds with return value 0
Press any key to continue . . .